Website at **www.kabellab.com**

Kabellab®

LabAir O²

Data Center Air-Conditioning Cooling System

Kabellab products are meticulously designed for easy installation while maintaining the highest quality with an aesthetically pleasing design. Kabellab combines innovative design with technology to create efficient products that achieve maximum impact. These are Kabellab's guiding principles.

1978 - 2017
IN LOVING MEMORY OF TLM

In-Room Precision Air-Conditioner

High Efficiency and Energy Saving

- Classified refrigeration system design, intelligent cooling capacity adjustment
- Stepless speed regulation EC fan (Optional underfloor design)
- High precision PID adjustment water valve (Chilled water type)
- High COP flexible scroll compressor
- Stepless speed regulation scythe type condensing fan
- Complete intelligent control system ensures dynamic optimization
- Optimum design of heat exchanger and air duct by CFD, high efficiency and low resistance for heat and mass transfer
- Pleated G4 coarse filter with big surface area, large capacity and low resistance
- High-efficient and green refrigerant R410A

- **Varieties of extended energy-saving function:**
  1. Fluorine pump / Glycol free cooling function
  2. Dual-cooling sources function
  3. High-efficient DC inverter compressor
  4. Use high-precision electronic expansion valve

High Reliability

1. Uninterrupted Operation Design:
This suggests the device is designed for continuous, uninterrupted operation, essential for environments requiring constant heating or humidification.

2. Strict Testing and Inspection of All Parts:
This indicates a strict quality control process where every component is thoroughly tested and inspected for high quality and reliability.

3. Safe and Reliable PTC Electric Heating:
PTC heaters are safe and reliable, self-regulating to prevent overheating and reducing fire hazards, ensuring consistent, risk-free heating.

4. Far-Infrared Humidification:
Far-infrared humidifiers emit rays that offer health benefits such as better blood circulation and reduced air bacteria, making them efficient and health-friendly.

5. Complete Protection Alarm and Professional Diagnosis Function:
These features provide advanced safety and self-diagnosis for issues like overheating, easing maintenance and repair.

Advanced Techniques

1. 150 Quality Management and Lean Production (TPS):
This refers to a quality management system with 150 standards, combined with lean production principles like the Toyota Production System, ensuring efficient, high-quality IT equipment production.

2. Manufacturing Techniques for IT Equipment:
This indicates specialized manufacturing processes for IT equipment, focusing on precision, durability, and compatibility with IT systems.

3. Fine and Decent Black Cabinet, Matching IT Room Perfectly:
This refers to the product's sleek, professional black design, ideal for blending into standard IT environments.

4. High-Strength Frame, Suitable for Sea, Land, and Air Transportation:
This feature highlights the product's durability and robust design, suitable for safe transportation by sea, land, or air, and protecting IT equipment from shipping stresses.

Intelligent Management

1. Interface design

The device features a 7" LCD touchscreen with a user-friendly interface for one-touch control, vibrant color displays of system components, and graphical trends of temperature and humidity data.

2. Protection alarm

This device features comprehensive auto protection and alarm functions, professional diagnostic capabilities, full parameter measurement and adjustment, automatic restart, water leakage detection, and lightning protection.

3. Teamwork control

Equipped with a standard R5485 communication interface and supporting the Modbus-RTU protocol for seamless integration and control of up to 32 units in a team configuration.

Convenient Maintenance

The system's architecture is meticulously crafted to ensure effortless maintenance, with full front access that simplifies any service needs. It boasts an innovative direct-connect design, integrating the fan and motor into a single unit, which not only streamlines the mechanism but also obviates the necessity for regular belt replacement—a common maintenance issue with traditional systems. Additionally, it incorporates a state-of-the-art far-infrared humidifier that operates without the need for ongoing maintenance, ensuring optimal humidity levels are maintained without additional labor or costs. This feature is particularly beneficial in environments where consistent humidity control is paramount for equipment efficiency and longevity.

Versatile Application Use

The unit boasts a versatile cooling capacity, suitable for a broad operational ambient temperature range from -40 to +55°C. It offers multiple airflow directions, various inlet and outlet piping configurations, and diverse power supply designs. The system can accommodate long pipe connections and significant height differentials. Its far-infrared humidifier is adaptable to different water conditions. The product is eco-friendly, adhering to ROHS and REACH standards, and has acquired CE, UL, and TUV certifications. It also promises quick and flexible responses to custom requirements.

Indoor Unit

An indoor precision aircon unit maintains specific indoor climates, crucial for sensitive environments like server rooms.

1	2	3	4	5
HM	070	A	X	P
HM	080	C	X	P

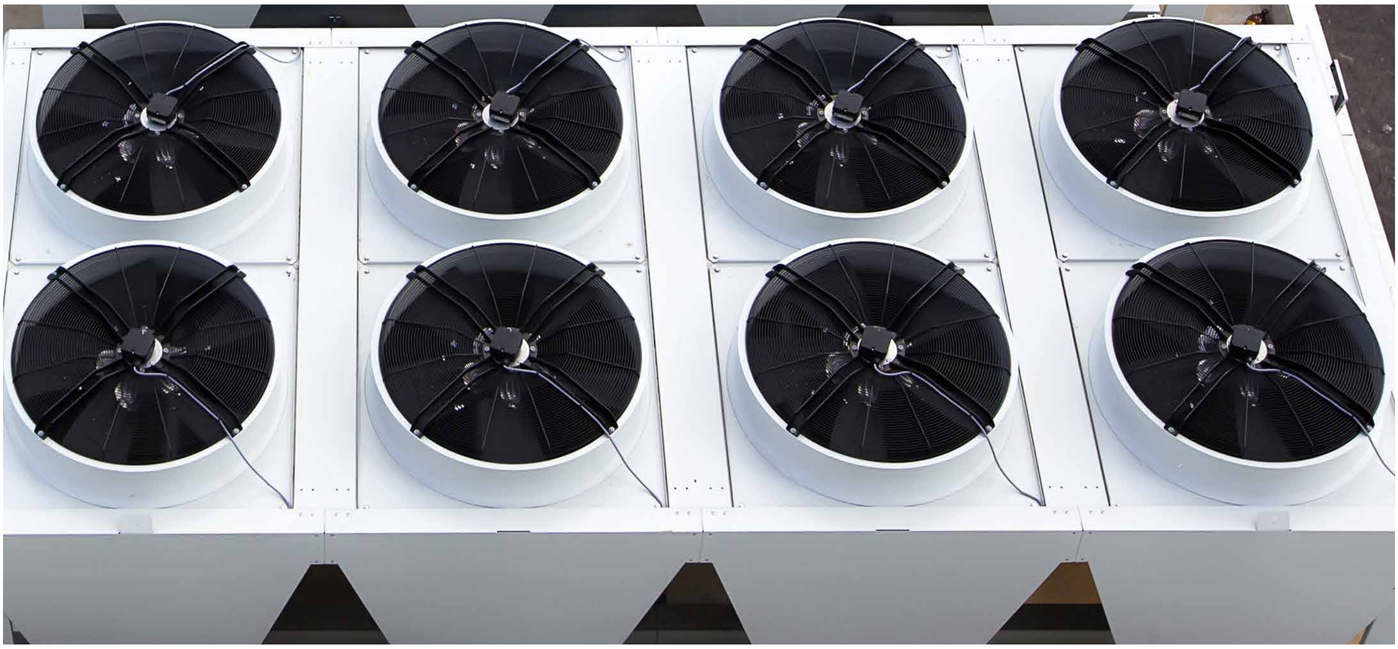
No	Item	Code	Description
1	Series	HM	Link-Wind Series air conditioner for data center
2	Cooling Capacity	070	Classification
3	Cooling Mode	A	air-cooled
		W	water-cooled
		C	chilled water
		--	others
4	Air Supply	X	bottom air supply by underfloor fan (standard)
		Q	forwards air supply by cap
		U	top air supply by duct
5	Function	B	cooling
		H	cooling and humidification
		P	cooling, heating and humidification

Outdoor Unit

A precision aircon outdoor unit precisely regulates external temperature and humidity for sensitive equipment.

1	2	3	4	5	6	7
SW	50	N	P	H	M	O

No	Item	Code	Description
1	Series	SW	outdoor unit for data center
2	Heat Exchange	50	for example, 50 - 50kw
3	Compressor Position	N	indoor (standard)
4	Heat Exchanger Shape	P	plate (standard)
		V	V shape
5	Refrigerant	H	R410A (standard)
6	Power Supply	M	3+PE~380V 50Hz
		Q	3+PE~380V 60Hz
		--	others
7	Type	0	normal temperature (standard)
		1	low temperature
		2	energy saving



Nominal working conditions and operation range

Chilled water In-Room Precision air conditioner

	Air Inlet		Water Inlet and Outlet	
Nominal Working Conditions	dry bulb temperature	wet bulb temperature	water inlet temperature	water outlet temperature
	24°C	17°C	7°C	12°C
Operation Range	indoor temperature		4~40°C	
	chilled-water inlet temperature		4~20°C	
	water side pressure		1.6MPa	
	altitude		≤1000, or derated	
	power adaptability		380V±10%, 50Hz±1Hz	

Note:

- **Overload operation for long time can cause damage. If there is any special requirement, please specify when placing order.**
- **Standard power supply is 380V/50HZ/3Ph. If there is any special requirement, please specify when placing order**

DX In-Room precision air conditioner

	Indoor		Outdoor		
Nominal Working Conditions	air inlet Status		air-cooled (air inlet Status)	water-cooled	
	dry bulb temperature	wet bulb temperature	dry bulb temperature	water inlet temperature	water outlet temperature
	24°C	17°C	35°C	30°C	35°C
Operation Range	indoor temperature			4~40°C	
	refrigeration system indoor temperature			18~32°C	
	air cooled	refrigeration system outdoor temperature		-15~+45°C	
	water cooled	cooling water inlet temperature		20~34°C	
		cooling water side pressure (plate heat exchanger)		≤1.0MPa	
	altitude			≤1000, or derated	
	power adaptability			380V±10%, 50Hz±1Hz	

Note:

- **Overload operation for long time can cause damage. If there is any special requirement, please specify when placing order.**
- **Standard power supply is 380V/50HZ/3Ph. If there is any special requirement, please specify when placing order**

Specifications

Chilled water In-Room Precision air conditioner

Item	HM	W040C	W050C	W060C	W070C	W080C	W100C	W120C	W130C	W150C	W180C	W200C
Air Flow	m³/h	8500	9500	11500	12500	17000	19000	23000	25000	28500	34500	37500
Cooling Capacity	kW	40.4	51.5	62	68.9	80.8	103	124	137.8	152	186	206.7
Sensible Cooling Capacity	kW	35.4	46	54.7	60.6	70.8	92	109.4	121.2	133.7	164.1	181.8
Water Flow	m³/h	6.9	8.8	10.6	11.8	13.9	17.7	21.3	23.6	26.1	31.9	35.4
Water Pressure Drop	kPa	82.4	82.4	70.4	84.8	72	72	71	71	78.6	82.4	96
electric heating capacity	kW	6	6	6	6	9	9	9	9	12	12	12
humidifying capacity	kg/h	4.5	4.5	4.5	4.5	6.5	6.5	6.5	6.5	10	10	10
Full Load Amps (FLA)	A	14.7	14.7	14.7	14.7	25.8	25.8	25.8	25.8	25.9	25.9	25.9
Dimension	width (mm)	900	900	900	900	1800	1800	1800	1800	2700	2700	2700
	depth (mm)	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
	height (mm)	1980	1980	1980	1980	1980	1980	1980	1980	1980	1980	1980
Weight	kg	300	320	325	330	560	600	610	620	880	905	920

Note:

- Operating conditions for return air: 24°C dry bulb and 17°C wet bulb temperatures.
- Chilled water supply and return temperatures: 7°C inlet and 12°C outlet.
- The device includes a standard two-way flow regulating valve for precise control of fluid flow. (Standard)
- The standard external static pressure is set at 50Pa for top discharge and 20Pa for down airflow configurations.
- If you require external static pressure beyond the standard levels, please reach out to us for assistance.
- Specifications may change due to product enhancements. Please verify before placing an order.

Indoor Unit Specifications

DX type In-Room precision air conditioner

Model	HM	K030	K045	K060	K090	K120
Indoor conditions: DB 35°C / RH 30%, Outdoor conditions: DB 35°C						
Cooling Capacity	kW	32.4	45.2	62.7	90.4	125.1
Sensible Cooling Capacity	kW	32.4	45.2	62.7	90.4	125.1
Indoor conditions: DB 24°C/RH 50%, Outdoor conditions: DB 35°C						
Cooling Capacity	kW	25.1	40.5	50.5	80.6	100.9
Sensible Cooling Capacity	kW	24.2	36.9	45.8	73.4	91.8
Air Flow	m^3/h	9000	11000	13000	22000	26000
Electric heating capacity	kW	6	9	9	9	9
Humidification volume	Kg/h	6	6	10	10	10
Maximum load current	A	49.7	62.2	69.2	110.8	124.8
Weight	kg	280	328	391	633	790
Dimension	width (mm)	1100	1100	1100	2200	2200
	depth (mm)	1000	1000	1000	1000	1000
	height (mm)	2000	2000	2000	2000	2000
Weight	kg	370	390	695	735	800
Flat panel outdoor unit specification						
Model		**50NP**	**60NP**	**80NP**	**60NP**	**80NP**
Matching quantity	Unit	1	1	1	2	2
Number of fans	Unit	1	2	2	4	4
Outdoor Unit Dimension	width (mm)	1650	2010	2700	2010	2700
	depth (mm)	1100	1100	1100	1100	1100
	height (mm)	1150	1150	1150	1150	1150

Note:

- Fully variable frequency scroll compressor, R410A environmentally friendly refrigerant.
- Power supply specifications: 380V, 3N, 50Hz.
- Working condition of water-cooled type: inlet and outlet water temperature 35/35°C
- Upward airflow room-level variable frequency units can be customized for a maximum external static pressure of 450Pa, maximum heating capacity of 36kW, and maximum humidification capacity of 15kg/h.
- For room-level variable frequency units, downward airflow defaults to a bottom outlet, and upward airflow defaults to a left outlet (compatible with bottom outlet).
- Specifications may change due to product enhancements. Please verify before placing an order.